

output remains high. Transistor T2 is in cut-off condition. Relay RL2 does not energise and hence IC2 does not toggle. As a result home appliance connected at the contacts of relay RL1 remains switched off.

When you press any remote key for the first time, IR receiver module's output goes low and collector of the transistor T2 goes high. Relay RL2 energises and triggers IC2. Output of

IC2 goes high and relay RL1 energises to switch on the appliance. Once relay RL1 is energised it remains in that state. So the appliance which is connected at the contacts of relay RL1 remains switched on.

Now when you press any remote key the second time, relay RL2 energises and re-triggers IC2. Output of IC2 goes low and relay RL1 de-energises to switch off the appliance. Once relay

RL1 de-energises it remains in that state. So the home appliance remains off. This cycle repeats when any key of the TV remote is pressed to switch on/off the home appliance.

Assemble the circuit on a general-purpose PCB and enclose in a suitable cabinet. Fix TSOP1738 and switch S1 on front side of the cabinet. Place transformer inside the cabinet and mains power cord at the back of the cabinet. ●

CCFL TESTER FOR LCD SCREENS

■ T.K. HAREENDRAN

Liquid crystal displays (LCDs) are used in a wide range of products including flat-screen computer monitors, laptop computers, tablet PCs, PDAs, digital cameras and portable instruments. Compact cold cathode fluorescent lamp (CCFL)-based backlight (illumination source for the LCD screen) arrangement in these applications enables a well-viewable display in both dim and bright ambient light conditions. LCD backlight mainly consists of a light source, diaphragm, light-leading board and plastic frame. It has high brightness, long lifetime and good uniformity features.

CCFL frequently employs fluorescent lamps, which have a phosphor-coated glass cylinder with cathodes at each end. CCFL tube is powered by a small electronic inverter (CCFL inverter) circuit that glows the screen electronically. The inverter circuit accepts a low-level DC input voltage and provides a high-level AC output to run the backlight CCFL tube(s).

Fig. 1 shows the CCFLs for LCD. Often a need arises to check a faulty LCD backlight circuit for finding the dead components. Absence of backlight means either the CCFL inverter and/or the CCFL tube is in dead state.

Here is a simple circuit to test the CCFL tube in a LCD backlight unit, which might help the hardware techni-

cian to speed up the repair work. This basic circuit (go/no go test) is portable, 6V battery operated and can be used to test almost all types of LCD backlight CCFL tubes.

The working of the CCFL circuit is shown in Fig. 2. A timer IC NE555 (IC1) is wired as an astable multivibra-



Fig. 1: CCFLs for LCD

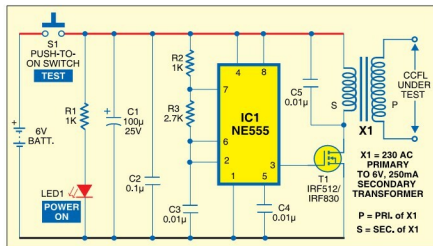


Fig. 2: Circuit diagram of CCFL tester



tor (AMV) to drive a standard MOSFET T1 (IRF 512/IRF 830). Components R2, R3 and C3 determine the operating frequency of the AMV. MOSFET T1

switches the inverter output transformer (X1) to produce high-voltage AC supply at its output terminals. Here an ordinary step-down transformer is used as the inverter transformer by reversing its primary and secondary windings.

Assemble the circuit on a general-purpose PCB and enclose in a suitable cabinet. Fix the 2-pin connector on front side of the cabinet in such a